Supplementary Material for PREFLOW: Open Delegation of Long-Horizon Web Tasks for Blind and Low-Vision Users

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About This Supplement

The main paper argues that a Web agent can be trusted with open delegation only if it maintains, outside the language model's context window, an explicit memory of what the user prefers and what the visited pages reveal about each preference. This supplement follows that argument from the mechanism to the people it is meant to serve, in seven sections. Table S1 indexes the questions that a reader is likely to ask and where each is answered.

Numbers without the prefix S, such as Section 3, Table 2, or Appendix A.3, refer to the main paper. The analyses added here reuse the runs and the study reported in the main paper. The anonymous repository holds the released code and a video demonstration of PREFLOW.

S1 Nora's Task, Step by Step

In brief. PREFLOW never relies on the language model to remember what Nora wants or what it has seen. Both live in an explicit memory that every later decision reads: what to ask, which desk to propose, and what to tell her before buying.

Figure S1 follows Nora's task (Section 1) through the six roles of Figure 1 and the steps of Algorithm 1. Table S2 shows the explicit memory after her revision, and Figure S2 shows the same task as Nora experiences it in the study interface (Section 5.3). The example is constructed, not recorded; Figure S3 in Section S3 follows a recorded run, and the video demonstration in the anonymous repository shows PREFLOW in use.

Three moments show why the memory must be explicit.

The question is grounded in observations (step 4). PREFLOW asks about price and assembly because the four desks it found differ in both, and the instruction leaves both open (DG4). It does not ask about properties that none of those desks exhibits, and, in this task, it does not ask before it has observed desks.

The revision is local (step 7). Lowering the budget changes one belief, the price ceiling, and keeps the others (DG1). The *Evidence Ledger* then rechecks every recorded desk against the new ceiling (Algorithm 1, line 6) instead of relying on the model to recall which desk cost what (DG2). Desk 1, the only desk that supported every preference, now exceeds the ceiling, so the recommendation must change.

The purchase summary names what is unmet (steps 8–9). After the revision, no desk supports every preference, and neither Desk 2 nor Desk 3 supports all that the other supports. The *Commit Verifier Agent* therefore does not choose on its own; the trade-off goes to Nora (Section 3, Remarks). The purchase summary is composed from the Evidence Ledger's statuses, so it states the one preference that the chosen desk contradicts, assembly (DG3, DG4). An agent that keeps preferences only in its context window has no record against which to check such a purchase summary.

Table S1: Questions a reader may ask, our short answers, and where this supplement supports them.

Question and short answer	Where
What does the explicit memory hold, and how does it change? Preference beliefs with their sources and, for every observed alternative, each belief's status with its evidence. A revision changes one belief and triggers a recheck of every recorded alternative.	S1; Table S2
Does PREFLOW see the benchmarks' goals or answers? No. No role reads WebShop's stored goals or rewards, or EComAgentBench's scripted answers; settings are identical for both LLMs; held-out tasks yield similar accuracy.	S2.3, S4.4
Why WebShop and EComAgentBench? Both pose open delegation, execute the agent's actions, and score each preference separately. WebShop has published results for the agents we rerun; EComAgentBench tags each preference with its source.	S3
Where does the WebShop gain come from? Mostly from the explicit memory, which improves all four reward components; the *Commit Verifier Agent* further improves option selection.	S4.1
Does the gain survive a weaker LLM? Yes. From Gemini-3-Flash to DeepSeek-V3.2, LASER loses 13.5–16.0 points on every reward component; PREFLOW loses at most 3.6.	S4.2
Does the memory matter more in long tasks? The observed gain is larger. In trajectories of more than 10 Web actions, it adds 9.0 points of complete satisfaction (7 instructions gained, none lost), versus 3.3 points otherwise.	S4.3
Are the differences beyond sampling noise? Yes for the paired comparison with LASER's policy on 500 instructions (+5.6 points, $p < 0.0001$); a 100-instruction sample alone is too small to tell.	S4.4
Is the EComAgentBench gain due to the LLM? No. With the same LLM, PREFLOW is correct in 25 of the 64 tasks in which the benchmark's own agent runs out of steps.	S4.5
Can a user who cannot see the page trust the report? Mostly. Before committing, the *Evidence Ledger* had flagged a doubtful preference in 83% of WebShop and 88% of EComAgentBench failures; silent failures occur in 6–8% of tasks.	S5; App. C
Does PREFLOW ask too much? At most four questions on EComAgentBench; nine of ten BLV participants rated the number of questions about right.	S4.5, S6.7
Did every BLV participant benefit? Trust was higher with PREFLOW for eight of ten and workload lower for seven, with individual reversals. The favorable mean changes remain without P3 and among the three non-daily chat-assistant users.	S6.6
Does the design extend beyond shopping, and what remains open? The roles and the control flow stay; the task schema, evidence rules, and commit checks change.	S7
How does PREFLOW differ from deployed conversational shopping assistants? Conversation and checkout are now shared capabilities; PREFLOW additionally maintains task-level preference beliefs linked to observed evidence and uses them for revisions, commitment checks, and reports.	S3.2

S2 Implementation Details

In brief. The same six roles run in all three settings; what changes is how each role observes and acts. No role reads a benchmark's stored goals, rewards, or scripted answers.

This section complements Appendix A. Section S2.1 explains the task schemas, Table S4 shows how each role is implemented in the three settings, and Section S2.2 gives the decision rules; Table S5 lists the key settings.

	Role	Step in Nora's task
1	*Task Router*	Nora asks: "I need a black folding desk." ROUTE returns transactional.obtain.
2	*Schema-Guided Planning Agent*	Preference beliefs: product *folding desk*; preference *color: black*; source: instruction.
3	*Executive Controller Agent*	Searches for folding desks and finds four: Desk 1 (black, fully assembled, $379.99), Desk 2 (black, requires assembly, $349.99), Desk 3 (white, fully assembled, $299.99), and Desk 4 (natural wood, requires assembly, $229.99).
4	*Dialogue Agent*	The four desks differ in price and in whether they arrive assembled, and the instruction leaves both open, so the agent asks: "What's your budget, and would you like it fully assembled?" Nora: "Under $400. I'd prefer it fully assembled." UPDATE adds *price: below $400* and *assembly: fully assembled*, both with source *reply*.
5	*Evidence Ledger*	Desk 1: supported {folding desk, black, price, fully assembled}. Desk 2: supported {folding desk, black, price}, contradicted {fully assembled}. Desk 3: supported {folding desk, fully assembled, price}, contradicted {black}. Desk 4: supported {folding desk, price}, contradicted {black, fully assembled}.
6	*Dialogue Agent*	Offers Desks 1–3 with what each meets and misses, and recommends Desk 1, which supports every preference. Nora asks what Desk 2 looks like and whether it is hard to assemble. The reply describes the photo ("The photo shows a matte black rectangular top on black metal legs…") and says what no observation addresses: "The listing doesn't say how hard assembly is."
7	*Schema-Guided Planning Agent*, *Evidence Ledger*	Nora: "My budget just changed. It's under $350 now." UPDATE revises the price ceiling, and the Evidence Ledger rechecks each desk's price: Desk 1 ($379.99) now contradicts it.
8	*Commit Verifier Agent*	No desk now supports every preference. Desks 2 and 3 each support one that the other does not, so the choice is a trade-off for Nora. The *Dialogue Agent* asks: "Which matters more to you: black, or fully assembled?" Nora: "Black. My brother can help me assemble it, so let's get the black office desk."
9	*Dialogue Agent*	Purchase summary ρ_{pre}: "Folding Office Desk (Black), quantity 1, $349.99. It's black, folds, and is within your $350 budget. It doesn't meet your preference for fully assembled: it needs assembly. Shall I buy it for $349.99?" Nora: "Yes, buy it."
10	Final action and report	The simulated purchase succeeds. Report ρ_F: "Done. I bought the Folding Office Desk (Black) for $349.99. It's a simulated purchase, so nothing was charged. Remember it needs assembly."

Figure S1: PREFLOW's roles in Nora's task (constructed example). Each row is one step of Algorithm 1 and names the role that performs it. Table S2 shows the explicit memory after step 7; Figure S2 shows the dialogue in the study interface.

Table S2: PREFLOW's explicit memory in Nora's task after her revision (step 7 in Figure S1; constructed example). Top: the preference beliefs $\widehat{P}$, each with its source. Bottom: the *Evidence Ledger* ℓ, with the status of each belief for each desk and the observation behind it.

Preference beliefs $\widehat{P}$

Belief	Value	Source	History
Product	folding desk	instruction	step 2
Color	black	instruction	step 2
Price	below $350	reply	below $400 at step 4; revised at step 7
Assembly	fully assembled	reply	step 4 ("I'd prefer")

Evidence Ledger ℓ (✓ supported, ✗ contradicted)

Desk	Product	Color	Price	Assembly
Desk 1	✓ title	✓ title	✗ $379.99	✓ "fully assembled"
Desk 2	✓ title	✓ title	✓ $349.99	✗ "requires assembly"
Desk 3	✓ title	✗ white	✓ $299.99	✓ "fully assembled"
Desk 4	✓ title	✗ natural wood	✓ $229.99	✗ "requires assembly"

After the revision, the Evidence Ledger rechecks all four prices against the new ceiling. Desk 1 now exceeds it, so no desk supports every preference, and the trade-off between color and assembly goes to Nora. Separately, the Evidence Ledger records what each desk's photo shows, such as Desk 2's black top and legs.

S2.1 Task Schemas

A Web page does not determine the task. The same page about Paris may serve a user who wants to learn the city's history and one who is planning a trip, and each needs something different from the agent. The *Task Router* therefore starts from what the user wants to do on the Web. Taxonomies of Web search intent supply the categories: Broder's informational, navigational, and transactional types [4] and Jansen et al.'s subtypes [19]. We found no taxonomy that specifies what the user needs from each category, so our task schemas include this: each schema lists the slots that the outcomes of its tasks require (Table S3). The history learner's task is informational, with a topic and an information need; the trip planner's is transactional, with the service, its attributes and options, and other preferences such as dates and budget. These slots are what the *Schema-Guided Planning Agent* fills and what the *Dialogue Agent*'s questions are about. A schema describes the intended outcome even when execution mixes types: a purchase also involves searching and navigating. Only transactional.obtain is evaluated (Appendix A.1); the other rows indicate how PREFLOW could extend to other tasks (Section S7).

Table S3: Task schemas and their slots. Informational subtypes share the slots shown. Only transactional.obtain is evaluated.

Type	Subtype	Slots
Informational	Directed, undirected, find, list, advice	Topic; information need; other preferences; source or format
Navigational	To an informational site	Target site
Navigational	To a transactional site	Target service or site; intended action
Transactional	Obtain	Product or service; attributes; options; other preferences
Transactional	Download	File or software; version; format; cost or license
Transactional	Interact	Service or action; required inputs; preferences

S2.2 Decision Rules

This section adds to Appendix A the rules that decide each step; their limits and thresholds are in the released code.

Planning. In WebShop, the *Evidence Ledger* classifies each preference by comparing it with product pages, and a price ceiling missing from the model's output is read from the instruction.

Executive Controller Agent. In WebShop, the controller follows LASER's policy [23]; invalid model outputs are repaired or retried. *Option handling* compares, before a purchase, the selected options with the preference beliefs for each option group, such as color or size. It selects a value only if it matches a preference and appears on the page, and it replaces the controller's selection only if the new value keeps the preferences the old one matched and matches more. *Recovery* starts from the controller's choice among the visited products, checks option conflicts, price, and product type against the beliefs, and may visit a few more products before buying the best one. In EComAgentBench, the controller retrieves the profile, runs the planned search queries (relaxing queries that return no results, and searching any store that the instruction names), ranks the results by how well their titles match the preferences, lets one

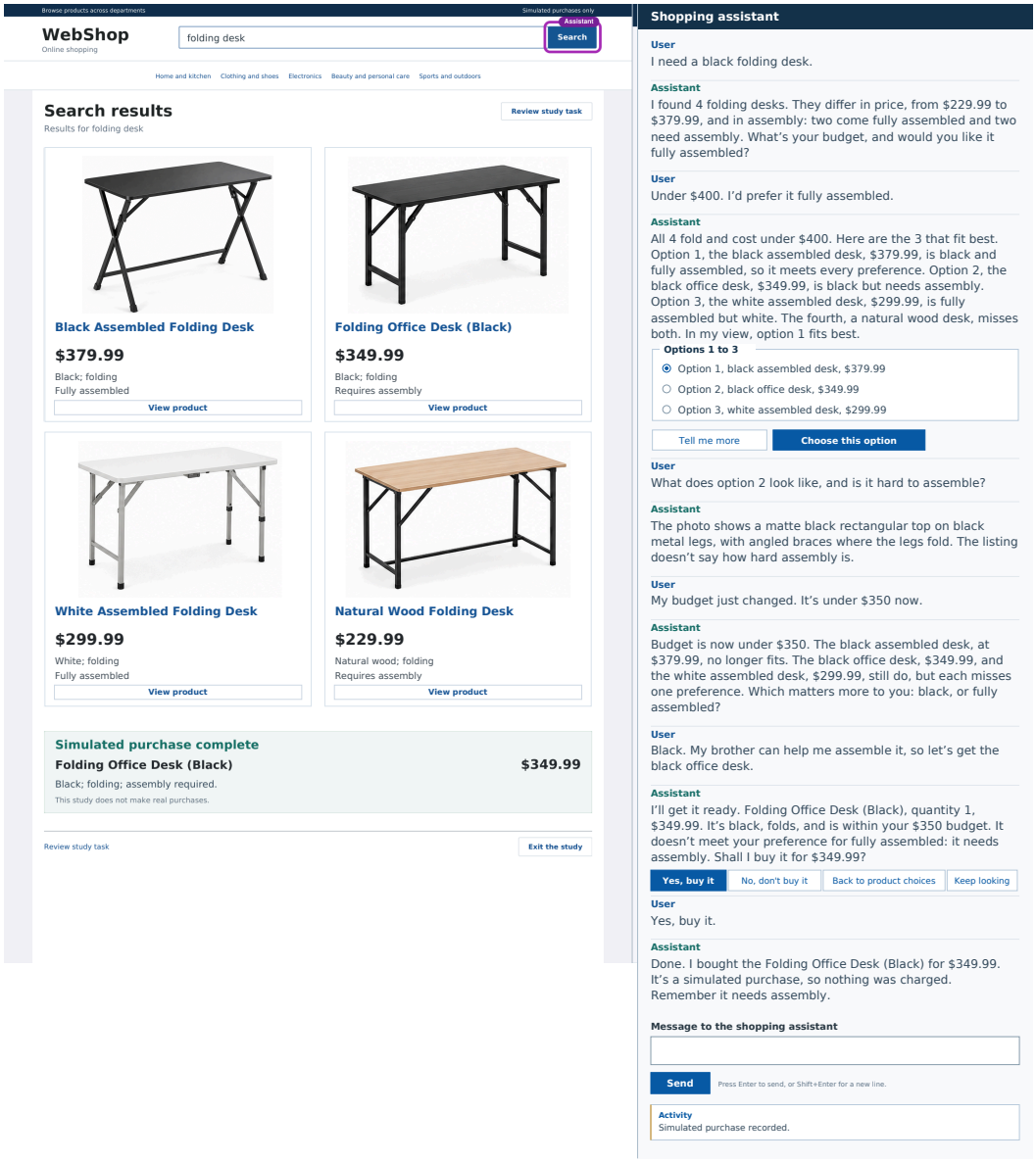


Figure S2: Nora's task in the study's Agent condition, with the shop beside PREFLOW's panel (scripted dialogue, not participant data). After finding four desks, PREFLOW asks about budget and assembly and recommends the desk that meets every preference. It describes another desk's photo and says what the listing does not state. When Nora lowers her budget, it rechecks the desks, presents the remaining trade-off, summarizes the purchase with the unmet preference, and buys only after explicit approval. Product images are generated illustrations; the purchase is simulated.

model call shortlist products for inspection, and, after the questions, searches again and inspects a few more top-ranked products.

Evidence Ledger. In WebShop, $M_A(y)$ in the score of Appendix A.3 contains the preferences that y's title or text matches, and $M_O(y)$ those that exactly one of its options matches. On a results page, where options are not yet visible, the option terms are dropped, and the type estimate is reduced when no preference matches the title or options. The ranking redirects the controller to a clearly better-ranked result, surfaces first-page results that the controller skipped and products from further result pages, and issues reformulated searches when a preference that can be checked in titles matches no visited product. In EComAgentBench, the Evidence Ledger keeps every tool result and a snapshot of each inspected product (title, store, price, details, features, description, rating statistics, and review excerpts); the verification call gives each status an evidence quote and a reason, and deterministic checks then mark a violated numeric requirement as contradicted and a status whose quote does not appear in the evidence as unresolved.

Table S4: Implementation of each component in the three settings. The BLV study adds the interactive parts of Algorithm 1 (lines marked †), which neither benchmark supplies.

Component	WebShop	EComAgentBench	BLV study
Task Router	transactional.obtain (all tasks are shopping)	transactional.obtain	transactional.obtain
Schema-Guided Planning Agent	One model call; preference beliefs with product, preferences, and price ceiling	One model call; preferences with their source; profile and answers add preferences	As in WebShop; replies add, revise, or withdraw preferences
Executive Controller Agent	LASER's policy over Web pages	Fixed stages over the benchmark's tools	As in WebShop; also searches further when the user asks for more options
Evidence Ledger	Matches between preferences and product pages; ranking and further search	Statuses with evidence quotes from the verification call, checked deterministically	As in WebShop; also records, separately, what product photos show
Commit Verifier Agent	Four deterministic checks before purchase	Model call assesses each preference; challenger check	Never buys on its own: purchase only after the user's approval, and never a product the request rules out
Dialogue Agent	Off (no user)	Up to four questions about the schema's slots, without a model call	Questions (some, such as size or budget, before the first search), trade-offs, photo descriptions, purchase summary and approval, report
Final action	Purchase	Recommendation	Simulated purchase after approval, or none

Table S5: Key settings; WebShop's are identical for both LLMs. All other limits and thresholds are in the released code.

Setting	Value
WebShop	
LLMs	Gemini-3-Flash; DeepSeek-V3.2
Temperature; top-p	0; 1
Output tokens: planning; controller	400; 200
Controller main-loop iterations	13
EComAgentBench	
LLM; temperature	Gemini-3-Flash; 1.0
Evaluator; temperature	Gemini-3.1-Pro; 1
Model calls per task	Up to 4
Questions per task	Up to 4
Products inspected per task	Up to 16
BLV study	
LLM (text and photos)	Gemini-3.7-Flash
Temperature	Provider default (not settable)

Commit Verifier Agent. In WebShop, the four checks run in a fixed order, and the product changes at most once per task. (1) *Other search results*: if the proposed product does not match every preference or lacks a required option, the Commit Verifier Agent visits a few other products on the current results page that are within the price ceiling and have at least the same type estimate, and replaces the product with the one whose page matches the most preferences, provided that it matches more than the proposed product and loses none of its matches; failing that, it may replace the product with one whose full-page text, including WebShop's repeated instruction, matches more preferences, under the same conditions and a score margin. (2) *Visited products*: it replaces the product with a visited one that matches every preference that the proposed product matches, and at least one more. (3) *Best-ranked product*: it replaces the product with the Evidence Ledger's top-ranked product if that product is within the price ceiling and either has a higher type estimate and a clearly higher score, or the same type estimate and more options that match preferences. (4) *Missing option*: if the proposed product lacks a required option, it replaces it with a recorded product that offers the option, has no conflicting option, is within the price ceiling, and has at least the same type estimate and a similar score. Checks 1 and 2 follow the rule of Section 4.4; the fallback of check 1 and checks 3 and 4 may lose a match elsewhere, within their conditions. In EComAgentBench, the challenger check runs only if a preference is not supported: it searches again for the unsupported preferences and inspects a few more top-ranked products, and a second model call compares the selection with all inspected products; its choice is kept unless it introduces a numeric contradiction that the selection did not have.

Dialogue Agent. In EComAgentBench, after the shortlisted products are inspected, Ask asks up to four short questions about the slots of transactional.obtain: usually first about the budget ("What is your budget for this item?"), then about attributes and other preferences, such as special features, ratings, or connectivity ("Do you need a specific connectivity technology?"). Figure S3 shows four.

S2.3 What PREFLOW Does Not See

An agent could gain simply by reading what the evaluator checks; PREFLOW cannot.

No stored goals or rewards. In WebShop, the *Schema-Guided Planning Agent* sees only the instruction, and the *Evidence Ledger* and the *Commit Verifier Agent* see only the preference beliefs and the pages. None of them reads WebShop's stored attributes, options, price ceiling, or reward (Appendix A.3). The Evidence Ledger's ranking score has the same structure as WebShop's reward because both express the notion of success in Section 3: the share of preferences that an outcome satisfies, gated by the product type. The score is computed solely from beliefs and observations.

No echo of the instruction. WebShop prints the instruction at the top of every page. The Evidence Ledger removes this header before matching, so a product cannot appear to satisfy a preference merely because the instruction is shown on its page. Only the fallback for the Commit Verifier Agent's first check uses the full page (Section S2.2); the header contains the instruction that every agent receives, not the stored goal.

No scripted answers. In EComAgentBench, PREFLOW obtains clarification preferences only by asking; it never reads the simulated user's script or the target product (Appendix A.5).

No per-LLM tuning. All limits and thresholds (Table S5 and the released code) are identical for Gemini-3-Flash and DeepSeek-V3.2, yet the mean-satisfaction gain over LASER is larger with DeepSeek-V3.2 (Section S4.2).

A held-out split. On 100 tasks from the 412-task held-out split, its accuracy is 45.0% (95% interval [35.6, 54.8]), consistent with its 48.3% on all 662 tasks ([44.6, 52.1]; Section S4.4).

S3 Choice of Benchmarks and a Recorded Run

In brief. RQ1 requires executable shopping tasks under open delegation, with an evaluator that checks each preference individually. WebShop provides this with fully stated preferences and published results for the agents we compare against; EComAgentBench provides it with partly stated preferences, each tagged with its source, and results for seven current agents.

S3.1 Criteria and Candidates

RQ1 requires open delegation, an executable Web environment or tools, automatic evaluation of the final outcome, long-horizon tasks expressed in natural language, separate checks for each requested property, and fully or partially stated preferences (Section 3). Table S6 compares the benchmarks we considered against these criteria. *Open delegation*: the user states the goal, and the agent chooses the actions [6]. RealWebAssist poses closed delegation instead: users give one instruction at a time, and each predicted action is scored against the recorded one [36]. *Dynamic interaction*: the agent's actions are executed and return new observations; recorded traces do not qualify. *Outcome evaluation*: the final answer or state is checked automatically against the task's goal, possibly by a model. *Multi-step*: tasks require several interdependent actions. *Natural-language tasks*: goals are stated in natural language, not as structured fields. *Property checks*: each requested property is checked separately, even if the checks are combined into one score.

Two considerations decided the choice. First, WebShop is the standard benchmark for shopping tasks with fully stated preferences, and it is the benchmark on which the five agents we rerun (Act, ReAct, ASH, AskAct, LASER) report their results, which allows both a published and a same-LLM comparison. It is not saturated: its human shoppers satisfy all preferences in 50.0% of test instructions (Table 3). Second, among the benchmarks with partly stated preferences that meet all other criteria, EComAgentBench tags each preference with its source (instruction, profile, or clarification answer) and reports results for seven current agents [9]. Source tags are what RQ1b measures: they distinguish the preferences the agent must obtain from those stated in the instruction. Evaluating PREFLOW on the other candidates in Table S6 is future work.

S3.2 Deployed Conversational Shopping Systems

In brief. Deployed shopping assistants increasingly combine natural-language product discovery with checkout or purchasing. This makes conversation and transaction execution alone insufficient to distinguish PREFLOW. The difference is in how the delegation is controlled: PREFLOW explicitly represents the user's current preferences, records what each observed product shows about each preference, revises this state when the user changes what they want, and uses it to decide when to ask, whether to commit, and what remains unsatisfied.

Table S7 compares PREFLOW with five deployed systems whose public descriptions cover both conversational shopping and actions toward purchase. The comparison concerns publicly documented behavior, not proprietary implementations. A dash therefore means that the public description does not establish the criterion, not that the system could not implement it. A ~ marks a related capability that is narrower than the criterion used for PREFLOW.

The columns refine the three challenges of Section 1 for systems that already converse and transact. *Elicit* means that the assistant proactively asks for a preference missing from the request, rather than merely accepting another constraint volunteered by the user. *Revise* means that the user can change a preference during the task. *Pref. state* is stronger: the current task has an explicit representation of the preferences that still apply, including additions, revisions, and withdrawals (DG1). *Evidence* means that observations about each candidate are related to those preferences as supported, contradicted, or unresolved (DG2). *Recheck* means that a preference revision causes previously observed candidates to be reassessed, rather than applying only to subsequent search. *Outcome check* requires the proposed outcome to be checked against the current preferences before the costly final action (DG3), and *Unmet report* requires contradicted or unresolved preferences to be stated to the user (DG4). These criteria concern the control of an open delegation, not whether a system can recommend a product.

Conversation and execution are shared capabilities. InnoSearch provides conversational product discovery and a standardized checkout, including a Shop by Phone interface through which a user can search, ask about products, manage a cart, and complete a purchase without using a screen [17, 18]. Alexa for Shopping supports conversational product research, comparisons, cart building, and purchases, including purchases triggered by a target price and, for eligible products outside Amazon, purchase on another merchant's site [2]. Google's shopping experiences combine conversational discovery and refinement in AI Mode with agentic checkout for eligible merchants; Gemini also supports shopping and eligible in-app purchases [12–14]. ChatGPT supports conversational shopping research, including clarification questions and refinement of constraints, and offers checkout for eligible products and merchants [24, 25]. Perplexity combines natural-language product search and shopping-specific follow-up questions with Instant Buy for supported merchants [27, 28]. Thus, neither natural-language shopping nor completing a transaction is by itself the capability studied here.

Preference state is different from personalization. Several deployed systems personalize recommendations. Alexa for Shopping can use shopping history and conversations, and Amazon lets users view, update, or remove information used for personalization [3]; ChatGPT shopping research can use past conversations and ChatGPT memory [25]; Perplexity describes personalization from past searches, interactions, and saved preferences [28]. These are useful forms of user modeling, but their public descriptions do not establish the task-level state used by PREFLOW: a set of current preference beliefs, each with its source and revision history, against which every later observation and decision can be checked. The distinction matters when a preference changes during execution. Remembering that a user generally likes a property is not the same operation as replacing a price ceiling for the current purchase while preserving the other constraints.

Refinement is different from reassessment. ChatGPT provides the closest documented behavior to PREFLOW's interactive refinement:

Table S6: Benchmarks considered for RQ1, as described in the cited publications through September 2026. A cross means that the criterion is not part of the published protocol, not that the environment cannot be extended. Bold rows are the benchmarks used in this paper.

Benchmark	Environment	Open delegation	Dynamic interaction	Outcome evaluation	Multi-step	Natural-language tasks	Property checks	Fully stated preferences	Partly stated preferences
FormWoB/QAWoB [32]	Cached Web	✓	✓	✓	✓	✗	✓	✓	✗
VirtualHome [29]	Household simulator	✓	✓	✗	✓	✓	✗	✓	✗
AndroidEnv [33]	Mobile apps	✓	✓	✓	✓	✗	✗	✗	✗
Mind2Web [8]	Recorded Web traces	✓	✗	✗	✓	✓	✗	✓	✗
WebArena [37]	Hosted Web	✓	✓	✓	✓	✓	✓	✓	✗
WebLINX [22]	Recorded Web dialogue	✓	✗	✗	✓	✓	✗	✗	✓
RealWebAssist [36]	Recorded Web dialogue	✗	✗	✗	✓	✓	✗	✗	✓
REAL [11]	Hosted Web	✓	✓	✓	✓	✓	✓	✓	✗
Mind2Web 2 [15]	Live Web research	✓	✓	✓	✓	✓	✓	✓	✗
AgenticShop [20]	Live Web	✓	✓	✓	✓	✓	✓	✓	✗
WebMall [26]	Hosted shops	✓	✓	✓	✓	✓	✗	✓	✗
ChatShop [7]	Search tools, dialogue	✓	✓	✓	✓	✓	✓	✓	✓
PROCLARE [35]	Retrieval, dialogue	✓	✓	✓	✓	✓	✗	✗	✓
UserBench [30]	Travel tools, dialogue	✓	✓	✓	✓	✓	✗	✗	✓
Persona2Web [21]	Live Web, history	✓	✓	✓	✓	✓	✓	✓	✓
ACWorld [10]	Commerce tools	✓	✓	✓	✓	✓	✓	✓	✓
WebShop [34]	Simulated shop	✓	✓	✓	✓	✓	✓	✓	✗
EComAgentBench [9]	Shopping tools, dialogue	✓	✓	✓	✓	✓	✓	✗	✓

Table S7: PREFLOW and deployed conversational shopping systems, as described in their public documentation. ✓: the documented behavior provides the criterion; ∼: related but narrower behavior; ✗: the criterion is not established by the public description. "Sim." denotes the simulated purchase used in PREFLOW's BLV study. The comparison concerns capabilities, not relative performance.

	Dialogue	Purchase	Elicit	Revise	Pref. state	Evidence	Recheck	Outcome check	Unmet report	Non-visual
InnoSearch [17, 18]	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓
Alexa for Shopping [1–3]	✓	✓	✓	∼	✗	✗	✗	∼	✗	✗
Google AI Mode / Gemini [12–14]	✓	✓	∼	✓	✗	✗	✗	∼	✗	✗
ChatGPT Shopping [24, 25]	✓	✓	✓	✓	✗	✗	∼	✗	∼	✗
Perplexity Shopping [27, 28]	✓	✓	✗	∼	✗	✗	✗	✗	✗	✗
PREFLOW	✓	Sim.	✓	✓	✓	✓	✓	✓	✓	✓

it asks clarification questions and lets the user change constraints, exclude products, and request alternatives while research is running [25]. Google AI Mode likewise supports continuous conversational refinement, including changing a product constraint in a follow-up query [12]. Amazon documents both follow-up questions for underspecified gift searches and editable personalization information, but not task-local revision semantics [1, 3]. Perplexity supports shopping-specific follow-up questions such as asking for a cheaper alternative [27]. We mark these related behaviors separately from *Recheck*. Public descriptions do not establish that a changed preference causes every relevant previously observed candidate to be checked again against the new value.

The distinction is visible in Nora's task (Section S1). When Nora changes her budget limit from under $400 to under $350, PREFLOW changes one preference belief and then rechecks the recorded desks against the new ceiling. The previously preferred Desk 1 changes from supporting the price preference to contradicting it; the remaining alternatives expose a trade-off between color and assembly. The revised preference therefore changes not only the next search query but the interpretation of evidence already collected. This operation follows from the explicit preference beliefs and Evidence Ledger rather than from the language model recalling the earlier product pages.

Checkout checks are narrower than preference checks. Deployed systems also use narrower safeguards around transactions. Google's agentic checkout can track a specific item down to size, color, and target price and buys only after the user confirms the purchase and shipping details [13]. Alexa for Shopping can monitor a target price and automatically purchase when that condition is met [2]. These mechanisms justify the partial entries under *Outcome check*, but they differ from DG3. PREFLOW checks the proposed outcome against the set of current preference beliefs, including properties learned from the instruction and later replies, and distinguishes a preference that is contradicted from one for which the observed pages provide no evidence. If no candidate dominates the others, the Commit Verifier Agent returns the trade-off to the user rather than resolving it silently.

Likewise, product comparisons and recommendation rationales may mention strengths or trade-offs. ChatGPT shopping research, for example, returns rationales, trade-offs, and comparisons against constraints supplied during the interaction [25]; this is why Table S7 marks its *Unmet report* capability as related but narrower. For PREFLOW, *Unmet report* specifically means that, before commitment, the purchase summary states which recorded preferences the proposed product does not satisfy or leaves unresolved. The final report again states the outcome and its unsatisfied preferences.

This is necessary for a user who cannot inspect the product pages to verify the agent's choice.

Non-visual shopping. InnoSearch is the closest deployed system in accessibility scope. Its Shop by Phone service explicitly supports searching, asking product questions, managing the cart, and purchasing via voice alone, and its Web interface is designed for screen reader use [17, 18]. PREFLOW should therefore not be distinguished by accessibility or voice interaction alone. Its contribution is complementary: the preference–evidence state supports questions, trade-offs, purchase summaries, and reports intended to let BLV users assess the delegated decision without revisiting the underlying pages. Section 6.3 evaluates this interaction with BLV participants.

Scope of the comparison. Table S7 is not an empirical comparison with these services. They are deployed products rather than reproducible benchmark agents, and their public documentation does not expose a common task set, internal state, or evaluator with which to measure preference satisfaction. They also have an important deployment advantage over the study system: InnoSearch, Amazon, Google, ChatGPT, and Perplexity support real transactions in at least some settings, whereas purchases in PREFLOW's BLV study were simulated. Our claim is therefore not that PREFLOW provides broader commerce infrastructure. The comparison isolates the capability addressed by the paper: maintaining an inspectable relation between evolving user preferences, observed evidence, and the decisions of a long-horizon delegate agent.

S3.3 A Recorded EComAgentBench Run

Figure S3 follows PREFLOW through one task, chosen because it shows every stage. An answer to a question introduces a preference that the first selection contradicts; the challenger check then finds a product that supports all nine preferences, and that product is the target.

S4 Where the Gains Come From, and How Far They Hold

In brief. The explicit memory, not the LLM, carries most of the gain. It improves every reward component, has a larger observed gain in long trajectories, and retains similar mean preference satisfaction under a weaker LLM. On EComAgentBench, PREFLOW's advantage over the benchmark's own agent stems from consistently reaching a recommendation that is checked against the recorded evidence.

S4.1 Which Preferences Does PREFLOW Satisfy More Often?

Figure 2 of the main paper decomposes WebShop's reward into its four components (Appendix B.1) for each agent and LLM on the 100 sampled instructions; Figure S4 does the same for the ablation on all 500.

With Gemini-3-Flash, PREFLOW exceeds LASER on product type (93.4% versus 91.8%) and options (67.7% versus 62.7%), ties it on price (99.0%), and is lower on attributes (80.7% versus 83.0%). On all 500 instructions, by contrast, the explicit memory raises attribute satisfaction over LASER's policy, from 76.4% to 80.0% (Figure S4). With DeepSeek-V3.2, PREFLOW is higher than LASER on all four components: options (64.1% versus 48.7%), price (98.0% versus 83.0%), product type (89.8% versus 76.5%), and attributes (82.3% versus 69.5%). For every agent and both LLMs, options have the lowest satisfaction.

Table S8: Change from Gemini-3-Flash to DeepSeek-V3.2 on the same 100 WebShop instructions, in percentage points (DeepSeek-V3.2 minus Gemini-3-Flash). Components from Figure 2; Mean and Complete from Table 2.

Agent	Type	Price	Option	Attribute	Mean	Complete
Act	−42.4	−47.0	−27.5	−39.0	−33.9	−16
ReAct	−36.0	−37.0	−22.4	−33.5	−30.3	−20
ASH	−11.1	−12.0	−5.7	−11.5	−9.1	−6
AskAct	−49.6	−52.0	−28.8	−47.0	−41.4	−27
LASER	−15.3	−16.0	−14.0	−13.5	−13.5	−6
PREFLOW	**−3.6**	**−1.0**	**−3.6**	**+1.6**	**−3.8**	−6

In the ablation, the explicit memory (V1) improves all four components over V0: type by 1.4 points, price by 1.0, options by 2.3, and attributes by 3.6. The *Commit Verifier Agent* (V2) improves only options, from 66.2% to 69.3%. Its role is narrow by design: it catches products whose required option, such as a size or color, is visible only on the product page.

S4.2 Does the Gain Survive a Weaker LLM?

Table S8 shows how much each agent loses on each reward component when the LLM changes from Gemini-3-Flash to DeepSeek-V3.2 on the same 100 instructions. Every other agent loses at least 5.7 points on every component, LASER between 13.5 and 16.0. PREFLOW loses at most 3.6, and its attribute satisfaction rises by 1.6.

The price component shows the mechanism most clearly. A price ceiling read in the instruction must still hold at the moment of purchase, often many pages later. With DeepSeek-V3.2, LASER exceeds the ceiling in 17% of instructions; PREFLOW, which stores the ceiling as a belief and checks it before purchase, exceeds it in 2%.

Complete satisfaction is 6 percentage points lower with DeepSeek-V3.2 for PREFLOW, LASER, and ASH (Table 2). Mean satisfaction drops by 3.8 points for PREFLOW and by 13.5 points for LASER, indicating greater stability in average partial satisfaction; these aggregates do not identify changes in individual instructions.

S4.3 Does the Memory Matter More in Long Tasks?

Section 6.1 reports a larger observed gain when V0 takes more than 10 Web actions, twice the median. Table S9 adds the complementary subset. Long trajectories are the hard ones: V0 satisfies all preferences in 28.2% of them, versus 54.3% of the others. There, the explicit memory adds 9.0 points without losing a single instruction; elsewhere, it adds 3.3 points. The within-subset tests favor V1 in both groups. The larger observed gain in long trajectories is descriptive; we do not report a test of the difference between subgroup gains.

Table S10 groups all 500 instructions by the number of preferences. Complete satisfaction is lower with four or more preferences (46.1–61.1%) than with three (68.6%), while mean satisfaction ranges from 73.1% to 81.8% across the groups.

	Stage	What happens
1	Task	Instruction: "I need a surveillance camera kit that comes with the 8 Bullet Cameras option and shoots in 1080p resolution. Ensure it includes 3 TB of storage, and only show me systems with at least 3 stars and at least 20 ratings." The profile requires a corded electric power source and compatibility with "Cameras LNAVA". Two further preferences are revealed only if asked.
2	*Schema-Guided Planning Agent*	Seven preferences: from the instruction, *8 bullet cameras, 1080p, 3 TB storage, at least 3 stars,* and *at least 20 ratings*; from the profile, *corded electric* and *compatible with Cameras LNAVA.*
3	*Executive Controller Agent*	Searches for camera kits and inspects the details, ratings, and reviews of seven.
4	*Dialogue Agent*	Asks four questions. "What is your budget for this item?" — "Hmm, I'm not sure about that." "Are there any special features you require?" — "I definitely need it to have night vision so I can keep an eye on things after dark." "Do you need a specific connectivity technology?" — "I want a wired system because I feel like it's a lot more reliable than relying on Wi-Fi." "Do you require a specific color?" — "Hmm, I'm not sure about that." Two answers add the preferences *night vision* and *wired*; a second search adds three more kits.
5	*Commit Verifier Agent*: selection	Selects the ZOSI 8CH 1080P Wireless Security Camera System. Eight preferences are supported, including night vision ("80ft Night Vision"); *wired* is contradicted ("Connectivity Technology: Wireless").
6	*Commit Verifier Agent*: challenger checks	Searches again for a wired system and inspects six more kits. The final model call compares all inspected kits and chooses the LaView 8 1080P IP Camera Security System, which supports all nine preferences ("Wired"; "3 TB"; "Cameras LNAVA"; "see up to 100 feet at night"). It introduces no numeric contradiction, so it replaces the ZOSI kit.
7	Recommendation and evaluation	PREFLOW recommends the LaView kit, which is the target product, after 4 questions, 16 inspected products, and 4 model calls.

Figure S3: PREFLOW on one EComAgentBench task with Gemini-3-Flash (recorded run of the reported version). Quotes are from the run's records; the user's answers come from the benchmark's script.

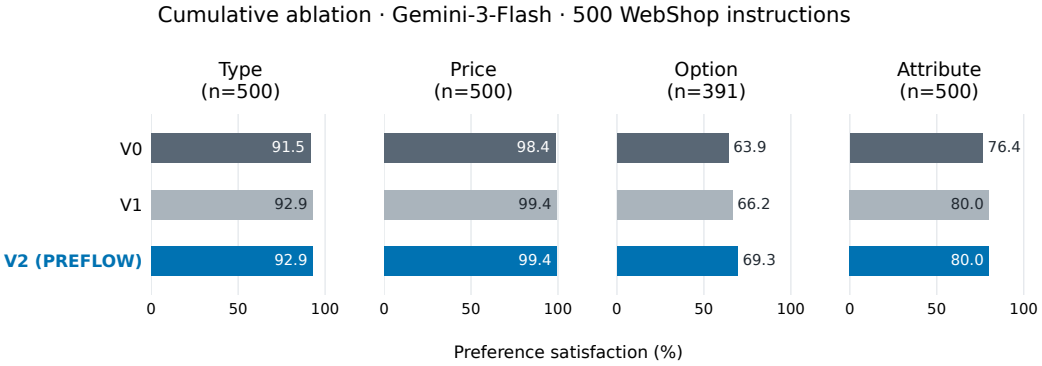


Figure S4: Satisfaction of each reward component in the cumulative ablation on the 500-instruction test split with Gemini-3-Flash. V0 is LASER's policy, V1 adds the explicit memory (preference beliefs and *Evidence Ledger*), and V2 adds the *Commit Verifier Agent*. Option satisfaction is averaged over the 391 instructions that require an option.

Table S9: Ablation V0 → V1 on the 500 WebShop instructions, split by the number of Web actions that V0 takes (median 5). Complete satisfaction in %; exact McNemar tests within each subset (Appendix B.2). Counts for the shorter subset follow from Table 8 and the longer subset.

V0's Web actions	N	V0	V1	Gained	Lost	p
At most 10	422	54.3	57.6	19	5	0.007
More than 10	78	28.2	37.2	7	0	0.016
All	500	50.2	54.4	26	5	0.0002

Table S10: PREFLOW's preference satisfaction by the number of preferences in the instruction, on all 500 WebShop test instructions (Gemini-3-Flash).

Preferences	N	Complete (%)	Mean (%)
3	70	68.6	79.4
4	208	61.1	81.8
5	167	46.1	73.1
6 or more	55	49.1	77.5

S4.4 How Large Is the Uncertainty?

WebShop runs use temperature 0 and EComAgentBench runs temperature 1.0 (Table S5); all intervals reflect variation across instructions or tasks, not across repeated runs (Appendix B.2). Table S11 gives 95% intervals for the main estimates, and two conclusions follow.

First, on 100 instructions a single proportion is uncertain by about ±10 points, so PREFLOW's 5-point lead over LASER on the 100 sampled instructions is not conclusive by itself. The paired comparison on all 500 instructions, in which V0 runs LASER's policy with the same LLM, supports a gain: +5.6 points [+3.0, +8.2], with 36 instructions gained and 8 lost ($p < 0.0001$; Table 8).

Second, on EComAgentBench, PREFLOW's accuracy point estimate is close to those of GPT-5.4 and Kimi K2.6 and below that of Claude Opus-4.6. These comparisons with published aggregates are descriptive; overlapping marginal intervals do not establish equivalence. What distinguishes PREFLOW is that it reaches this accuracy with 3.7 model calls per task instead of 25.1–38.2 for these three agents (Table 5).

Table S11: Main estimates with 95% intervals. Single proportions: Wilson intervals, computed from the reported percentages for published results. Paired differences: bootstrap intervals and exact McNemar tests (Appendix B.2).

Estimate	Value (%)	95% interval
WebShop, complete satisfaction		
PREFLOW, 100 instructions, Gemini-3-Flash	55.0	[45.2, 64.4]
LASER, 100 instructions, Gemini-3-Flash	50.0	[40.4, 59.6]
PREFLOW, 100 instructions, DeepSeek-V3.2	49.0	[39.4, 58.7]
LASER, 100 instructions, DeepSeek-V3.2	44.0	[34.7, 53.8]
PREFLOW, 500 instructions	55.8	[51.4, 60.1]
V0 (LASER's policy) → PREFLOW, 500, paired	+5.6	[+3.0, +8.2]
EComAgentBench, accuracy		
PREFLOW, 662 tasks	48.3	[44.6, 52.1]
GPT-5.4, 662 tasks	47.0	[43.2, 50.8]
Kimi K2.6, 662 tasks	46.4	[42.6, 50.2]
Claude Opus-4.6, 662 tasks	57.1	[53.3, 60.8]
PREFLOW, 100 held-out tasks	45.0	[35.6, 54.8]
Benchmark agent, 100 held-out tasks, same LLM	23.0	[15.8, 32.2]

Table S12: Correct recommendations on the 100 held-out EComAgentBench tasks with the same LLM (Gemini-3-Flash), split by whether the benchmark's agent finishes within its 100 steps. Derived from Section 6.2.

		Correct	
Tasks in which the benchmark's agent	N	Benchmark agent	PREFLOW
finishes	36	23	20
runs out of steps	64	0	25
All tasks	100	23	45

S4.5 Where Does the EComAgentBench Gain Come From?

With the same LLM on the same 100 held-out tasks, PREFLOW is correct in 45 tasks and the benchmark's own agent in 23 (Table 5; McNemar $p < 0.001$). Table S12 shows where the difference arises. The benchmark's agent exhausts its 100 steps in 64 tasks—in 18 of them after it had already retrieved the target product—and is correct in none of them. PREFLOW is correct in 25 of these 64. On the 36 tasks that the benchmark's agent completes, it is correct slightly more often (23 versus 20). PREFLOW's gain is therefore one of reliability: its fixed stages always reach a recommendation checked against the recorded evidence, with 21 times fewer model calls.

How broadly does PREFLOW search? Table S13 compares the search effort of both agents on the same 100 tasks. The benchmark's agent rarely looks beyond the first page of results and spends most of its steps on review searches; PREFLOW looks beyond the first page in almost every task and inspects three times as many products with fewer tool calls. Looking further matters: on all 662 tasks, 20 of PREFLOW's 320 correct recommendations ranked below the first results page in every search in which they appeared, so an agent that reads only first pages would not have seen them. Products found that deep are harder to judge: 20 of the 94 such recommendations were correct, against 258 of the 515 found on a first page.

Asking after observing also pays off. PREFLOW asks 4.0 questions per task, the most of any agent and its maximum, and satisfies the largest share of clarification preferences (72.2%; Table 5). With the same LLM, the benchmark's agent asks 0.2 questions per task and satisfies 27.0% of clarification preferences, against 66.5% for PREFLOW. Each task has two clarification preferences, which the simulated user reveals only when asked, so PREFLOW obtains them by asking.

Table S13: Search effort on the 100 held-out EComAgentBench tasks with the same LLM (Gemini-3-Flash).

	PREFLOW	Benchmark agent
Tasks with a result page beyond the first	96	4
Products inspected (median)	15	5
Review searches (median)	11	54.5
Tool calls (mean)	49.8	75.2
Model calls (mean)	3.6	75.2
Tasks ending at the step limit	0	64

Table S14: EComAgentBench accuracy (%) on all 662 tasks by the number of requirements that the instruction does not state. The average over the seven published agents and GPT-5.4's values are from [9]. Comparisons with the published aggregate rates are descriptive.

Unstated requirements	Tasks	Published average	GPT-5.4	PREFLOW
0–1	215	51.8	59.1	59.1
2	335	36.1	44.5	46.3
3 or more	112	27.2	31.2	33.9

Table S14 breaks down accuracy on all 662 tasks by the number of requirements that the instruction does not state, using the groups of the benchmark's own analysis [9]. PREFLOW ties GPT-5.4 when at most one requirement is unstated and leads by 1.8 and 2.7 points with two and with three or more.

S5 Can a User Trust What PREFLOW Reports?

In brief. Before committing, the *Evidence Ledger* knows which preferences it could not confirm. It had flagged a contradictory or unresolved preference in 83% of WebShop failures and 88% of EComAgentBench failures, so a report could have warned the user; 7.6% and 6.2% of tasks ended in silent failures.

For a user who cannot see the page, what the agent says about a purchase matters as much as the purchase. PREFLOW's purchase summary and report state the Evidence Ledger's statuses at commitment (Section 4.5), so two questions decide their value: when the Evidence Ledger supports every preference, is the outcome right; and when the outcome is wrong, had the Evidence Ledger flagged it?

Neither benchmark shows the report to a user, but the Evidence Ledger's state at the time of commitment is recorded. Figure 3 of the main paper crosses this state with each benchmark's evaluation: (a) all 500 WebShop test instructions (full PREFLOW, Gemini-3-Flash; 55.8% complete, as in Table 4) and (b) all 662 EComAgentBench tasks (48.3% correct, as in Table 5), where a model call assigns the statuses (Appendix A.3).

Full support is informative. When the Evidence Ledger supports every preference belief, 107 of 145 WebShop purchases (73.8%) satisfy all stored preferences, against 172 of the other 355 (48.5%). On EComAgentBench, 217 of 258 such recommendations (84.1%) are correct, against 103 of the other 404 (25.5%).

Most failures are flagged before committing. The Evidence Ledger had recorded a contradicted or unresolved preference in 183 of the 221 WebShop purchases that miss a stored preference (82.8%) and in 301 of the 342 incorrect EComAgentBench recommendations (88.0%). A report would have indicated which preference was in doubt.

Failures despite full support. 38 WebShop tasks (7.6%) and 41 EComAgentBench tasks (6.2%) end in a failure although the Evidence Ledger supported every belief.

String matching errs on the side of caution. Of the 279 complete WebShop purchases, 172 (61.6%) were not fully supported at commitment. The WebShop Evidence Ledger matches preferences against page text (Appendix A.3), so a preference remains unresolved when the page does not state it in matching words, even if the product has it. The model-based Evidence Ledger in EComAgentBench is less cautious: 103 of the 320 correct recommendations (32.2%) were not fully supported.

Silent failures point back to the formalization of Section 3. The Evidence Ledger checks observations against the beliefs $\widehat{P}$, not against the user's preferences P, so it cannot flag a preference that the beliefs omit or misread. A report can be no more accurate than the beliefs it is checked against, which is why PREFLOW records each belief along with its source and allows the user to correct it (DG1).

An agent that keeps preferences only in its context window, such as LASER, commits without a per-preference record and has nothing against which a report could be checked. For BLV users, checking product details with a screen reader can require revisiting several pages. A per-preference record can support a report that states what remains in doubt alongside what was bought.

S6 The BLV Study in Detail

In brief. Ten BLV participants each shopped once on their own and once with PREFLOW. Trust was higher with PREFLOW for eight, workload lower for seven, and seven preferred it; three had no preference. Nine rated the question count about right. Participants valued relevant options, product details, comparisons, and reduced screen-reader navigation, while some wanted different amounts of assistance.

This section gives the materials and full results of the study in Section 5.3; Appendix B.4 reports the ethics approval and consent.

S6.1 Participants and Procedure

Participants were recruited through organizations of blind and low-vision people (names omitted for anonymous review). Ten adults who are blind or have low vision and use a screen reader took part (Table S15). They received compensation by bank transfer as a thank-you for their time. They were told that the study evaluates two ways of shopping, not them; that there are no right or wrong answers; and that honest criticism is welcome. Interaction logs and questionnaire responses are stored under participant codes. An intake questionnaire records age range, vision status and its duration, assistive technology and experience with it, browser, online shopping frequency, and prior use of voice and chat assistants.

Table S16 lists the session steps. Condition order and task assignment used four sequences: Manual beach then Agent cold-city (four participants), Manual cold-city then Agent beach (two), Agent beach then Manual cold-city (two), and Agent cold-city then Manual beach (two). Thus, six completed Manual first and four Agent first. Sessions were held individually online on a video-call platform of the participant's choice, with the moderator present throughout. A short, unmeasured practice with PREFLOW precedes the Agent task.

Table S15: Participants. Proficiency with their screen reader is self-rated (1–5); chat assistants: how often they use conversational AI assistants.

	Age	Vision (duration)	Screen reader	Prof.	Chat assistants
P1	25–34	Blind, light perception (since birth)	NVDA	5	Daily
P2	25–34	Blind, light perception (since birth)	NVDA	5	Daily
P3	45–54	Blind (over 20 years)	JAWS	5	Daily
P4	25–34	Blind (since birth)	NVDA	5	Daily
P5	45–54	Blind, light perception, peripheral loss (11–20 years)	JAWS, braille display	4	Tried once
P6	35–44	Low vision, central loss (6–10 years)	VoiceOver, magnifier	3	Occasionally
P7	55–64	Blind (over 20 years)	JAWS	4	Never
P8	18–24	Blind, light perception (since birth)	VoiceOver, dictation	5	Daily
P9	25–34	Blind (6–10 years)	VoiceOver, dictation	5	Daily
P10	35–44	Blind (11–20 years)	NVDA	3	Daily

Table S16: Steps of a study session.

	Step	Content
1	Consent and intake	Informed consent; intake questionnaire
2	Orientation	Introduction to the shop with the participant's own assistive technology
3	First task	Manual or Agent condition; a short practice with PREFLOW precedes the Agent task
4	Questionnaire	SEQ, NASA-TLX, control, SUS, trust, product satisfaction; final preferences
5	Break	
6	Second task	The other condition and the other task
7	Questionnaire	As in step 4
8	End of session	Message items, preferred condition, open questions

S6.2 Interfaces

Both conditions use the same shop, built on WebShop's product catalog and search engine [34] (Figure S5). A search returns up to 50 products in pages of 10. Each product has a fixed purchase price that is the same in both conditions, and purchases are simulated. In the Manual condition, participants search, open product pages, select options, and buy on their own. The Agent condition adds PREFLOW's conversation panel next to the shop, and participants can still browse the shop directly. In this panel, the *Dialogue Agent* asks for missing preferences, presents trade-offs, requests approval, and reports the outcome. Table S17 lists how its interface serves screen-reader and low-vision users. The purchase summary ρ_{pre} states the product, selected options, exact price, and supported, contradicted, and unresolved preferences. Approval applies only to this product, these options, and this price; any later change cancels it. Figure S2 shows a scripted conversation in this interface.

Both conditions load the same shop pages, so every accessibility fix applies to both, such as a labeled search box, results named by product title rather than product code, ratings presented in text, and no meaning conveyed by color alone. The shop's ordinary friction

Table S17: How the Agent condition's interface serves screen-reader and low-vision users (target: WCAG 2.2 AA).

Need	What the interface does
Hear replies without losing one's place	Each reply is announced once without moving keyboard focus, with an optional sound; each message is a heading, so the latest reply is one keystroke away.
Know what PREFLOW is doing	A status region gives a progress line at least every 5 s and reports problems; an activity list records searches, products opened, and options selected, never the model's reasoning.
Stay in charge of the shop	The panel is a landmark placed before the shop. Pages that PREFLOW opens replace the shop area without reloading. Participants can browse or select options themselves, and PREFLOW continues from that state; replies link the products they mention.
Answer quickly	A question offers up to seven reply buttons, each sending the same text that the participant could type.
Compare products	Comparisons are spoken and also given as a data table with row and column headers.
Know what products look like	PREFLOW describes product photos on request and marks what comes from a photo (Appendix A.3).
Low vision	The layout reflows to one column; text contrast is at least 7:1; focus is clearly visible; forced colors and reduced motion are respected.

is kept as the baseline: the same ranking, ten results per page, no filters, and product information on separate sub-pages. PREFLOW uses the same search function, so rankings do not differ between conditions.

S6.3 Tasks and Revision

Both tasks are open: a task card suggests a shopping occasion for participants who have nothing in mind, and participants may seek any product in the shop. Participants form their own preferences and may change them at any time. For five participants (P4–P8), the researcher said midway through both tasks: "Imagine your budget has just changed. Your new limit is $\$X$. Please continue with that in mind." The catalog may not contain a requested product; participants can knowingly accept the closest available alternative. For example, P3 knew that Lenovo products were unavailable and accepted the closest available Dell-related option. A task ends with a purchase or when the participant ends it without buying.

S6.4 Questionnaires

The questionnaires were built for screen readers: each rating item is one labeled group that names both scale ends, there is no time limit, and every page offers an exit.

After each task. Participants rate the task's ease with the Single Ease Question [31] (1, very difficult, to 7, very easy) and workload on the six NASA-TLX dimensions [16]: mental demand, physical demand, temporal demand, performance, effort, and frustration, each from 0 to 100; the workload score is their unweighted mean (raw TLX). They then answer the following items, where *{interface}* is replaced by "shopping on my own" in the Manual condition and "the shopping assistant" in the Agent condition.

Control (1–7, strongly disagree to strongly agree):

(1) I was in control of what happened while I was using {interface}.
(2) I always knew what {interface} was doing.
(3) I could change course whenever I wanted to while using {interface}.

Table S18: End-of-session items about PREFLOW's messages, each rated from 1 (low anchor) to 5 (high anchor).

Item	Low / middle / high anchors
How many things did PREFLOW ask you about?	Far too few / About the right number / Far too many
How long were PREFLOW's replies?	Much too short / About the right length / Much too long

System Usability Scale [5] (1–5), with "awkward" instead of the original "cumbersome" in item 8:

(1) I think that I would like to use {interface} frequently.
(2) I found {interface} unnecessarily complex.
(3) I thought {interface} was easy to use.
(4) I think I would need support from a technical person to use {interface}.
(5) I found the various functions in {interface} were well integrated.
(6) I thought there was too much inconsistency in {interface}.
(7) I would imagine that most people would learn to use {interface} very quickly.
(8) I found {interface} very awkward to use.
(9) I felt very confident using {interface}.
(10) I needed to learn a lot before I could get started with {interface}.

SUS scores range from 0 to 100: each odd-numbered response minus 1 and 5 minus each even-numbered response are summed and multiplied by 2.5 [5].

Trust (1–7, strongly disagree to strongly agree):

(1) I could rely on {interface} to help me find what I wanted.
(2) I would trust {interface} to shop with again.

Product satisfaction. After a purchase, participants rate their satisfaction with the purchased product (1–7).

Final preferences. Participants list the preferences they had at the end of the task and mark, for each, whether they believe the purchased product satisfies it (yes, no, or not sure).

Additional items were introduced with P4 (Table S20). The five information and purchase-check items are available for P4–P10. The two change items have five complete pairs (P4–P8); P9 answered not applicable in both conditions, and P10 did so in Manual but rated both Agent items as 6.

End of session. Participants answer the two items in Table S18 about PREFLOW's messages. They then state whether they prefer shopping with PREFLOW, shopping on their own, or have no preference, explain why, and answer four open questions: "What was the most difficult part of today?", "What worked well for you?", "How sure were you that you bought the right thing?", and "If you could change one thing, what would it be?"

Records and analysis. Responses and interaction records were collected using automatic logging and manual transcription from full session recordings as needed. Some timestamps were unreliable, so no task-duration or response-latency comparisons are reported. We report per-condition means and sample standard deviations and each participant's paired differences descriptively. Trust and control are their item means. For each purchased task, the share of final preferences believed satisfied is the number marked yes

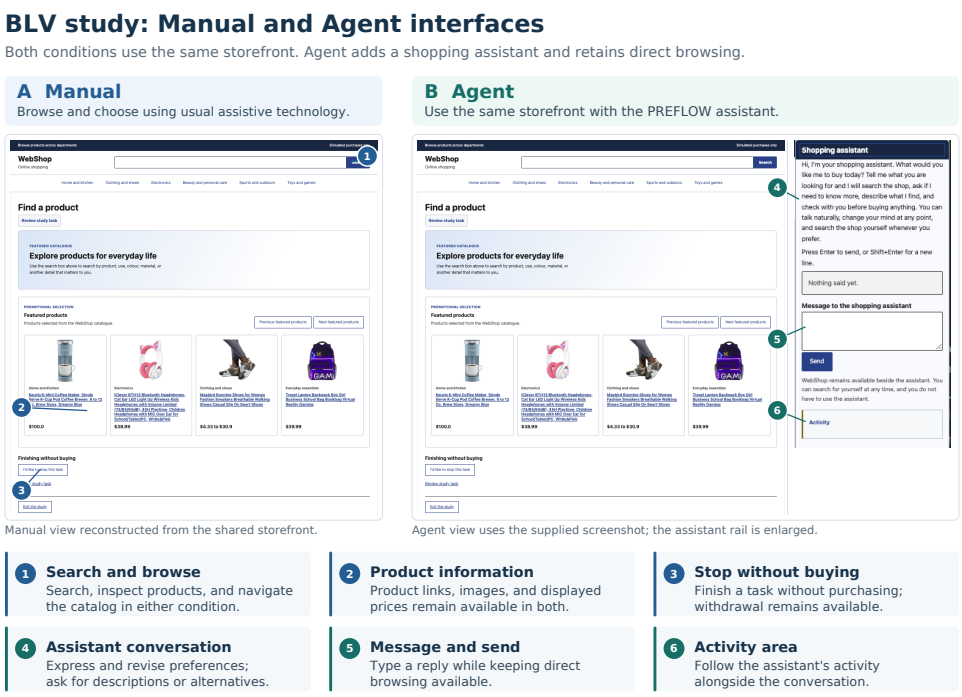


Figure S5: The Manual and Agent conditions use the same shop. The Agent condition adds PREFLOW's conversation panel, shown enlarged for legibility. Both conditions allow direct browsing and task completion without making a purchase. Purchases are simulated.

divided by all listed preferences (including no and not sure), then averaged equally across participants. Additional-item comparisons use complete pairs per item; not-applicable responses are excluded rather than set to zero. Purchases count completion in the condition, which also allows direct participant actions. Interaction records give question counts and purchase summaries, excluding practice. We translated Arabic answers into English; one author grouped open answers by topic, including P9's recording-based comment on filtering irrelevant products.

S6.5 Results

Table 6 of the main paper and Figure S6 give the means over ten sessions; Table S19 and Figure S7 give individual differences. Every participant bought a product in each condition. For P4–P8, all five purchases in each condition remained within the revised budget as prompted. These are condition-level purchases, not a count of purchases executed solely by the agent.

For the seven paired respondents to the information items (Table S20), ratings were higher with PREFLOW for finding out product appearance (5.7 versus 4.1) and having enough information to decide (5.7 versus 4.4). Perceived correctness was 5.9 versus 5.4; these ratings do not independently establish product correctness. In the five complete revision pairs, mean ratings favored PREFLOW for continuing after a change (6.2 versus 5.4) and for how well the products fit the change (6.4 versus 5.8).

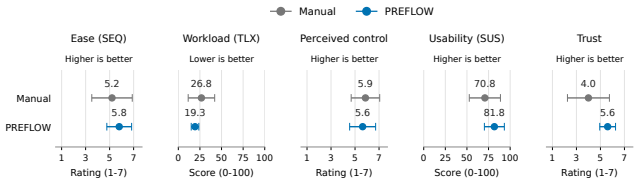


Figure S6: Mean ratings over ten participants in Manual and with PREFLOW; error bars show one sample standard deviation. Higher is better except for workload. Panels retain each measure's own scale.

S6.6 Who Benefited?

Means over ten participants hide individual differences (Table S19; Figure S7).

Direction. Trust was higher with PREFLOW for eight participants, equal for P8, and lower for P10. Workload was lower for seven, equal for P2, and higher for P1 and P10. Usability was higher for eight and lower for P1 and P2. Ease was higher for four, equal for five, and lower for P2.

Sensitivity and medians. P3 had the largest workload and ease changes; P3 and P9 tied for the largest usability (+42.5) and trust (+4.5) changes. Without P3, mean differences remain favorable for

Table S19: Each participant's difference, PREFLOW minus Manual; mean and median over ten participants and mean without P3. P3 has the largest workload and ease changes and ties P9 on usability and trust. Positive favors PREFLOW for ease, usability, and trust; negative for workload.

	Ease (SEQ)	Workload (TLX)	Usability (SUS)	Trust
P1	0	+9.2	−2.5	+1.5
P2	−3	0.0	−7.5	+2.0
P3	+5	−36.7	+42.5	+4.5
P4	+1	−12.5	+10.0	+1.0
P5	+2	−11.7	+10.0	+1.0
P6	0	−9.2	+5.0	+1.0
P7	+1	−9.2	+2.5	+1.0
P8	0	−1.7	+5.0	0.0
P9	0	−7.5	+42.5	+4.5
P10	0	+5.0	+2.5	−0.5
Mean, all ten	+0.6	−7.4	+11.0	+1.6
Median, all ten	0.0	−8.3	+5.0	+1.0
Mean without P3	+0.1	−4.2	+7.5	+1.3

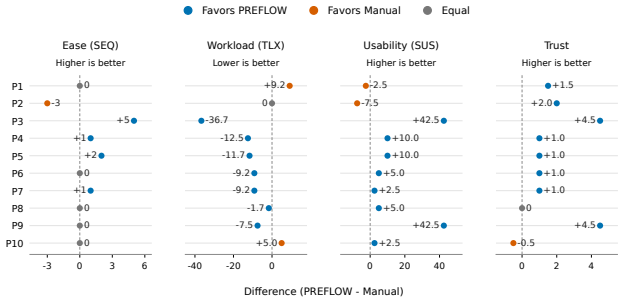


Figure S7: Each of ten participants' differences, PREFLOW minus Manual. Positive favors PREFLOW for ease, usability, and trust; negative favors it for workload. Zero denotes equal ratings.

Table S20: Additional items: complete-pair means (1–7; final item 1–5, with 3 "about right"). Change items use P4–P8; other items P4–P10. P10's Agent-only change ratings are excluded from paired means.

Item	Paired N	Manual	PREFLOW
After I changed what I wanted, it was easy to go on	5	5.4	6.2
The products offered or found fit the change	5	5.8	6.4
I had enough information about the products to decide	7	4.4	5.7
I could find out what the products looked like	7	4.1	5.7
The information was correct, as far as I could tell	7	5.4	5.9
Before the purchase I knew exactly which product, options and price	7	6.3	6.7
The checks before the purchase were (1 far too few, 5 far too many)	7	2.7	3.0

workload (−4.2), usability (+7.5), and trust (+1.3), with little difference in ease (+0.1). Across all ten participants, median differences are −8.3 for workload, +5.0 for usability, +1.0 for trust, and 0 for ease. The mean share of final preferences believed satisfied rises by 12.7 percentage points, but by only 3.0 points without P3; three participants report a higher share, five the same share, and two a lower share.

Beyond daily chat-assistant users. P5 had tried a chat assistant once, P6 used them occasionally, and P7 had never used one (Table S15); all three reported lower workload, higher usability, and higher trust with PREFLOW. P1 and P2, whose usability ratings were lower with PREFLOW, use chat assistants daily and rate their screen-reader proficiency at the maximum; their trust ratings were higher (+1.5, +2.0). These descriptive patterns do not establish an effect of prior experience.

Variation across participants. Standard deviations were lower with PREFLOW for workload (4.4 versus 15.3), trust (0.7 versus 1.7), usability (11.6 versus 18.0), and the share of final preferences believed satisfied (0.15 versus 0.37; Table 6). These summarize variation in this sample, not evidence of population-level uniformity.

The ten-person, single-session study is reported descriptively.

S6.7 Did PREFLOW Ask Too Much, and Did Participants Keep Control?

Questions. Measured Agent tasks contain an average of 10.0 recorded question events and 10.8 participant messages, excluding practice. Besides eliciting preferences, questions present trade-offs and request approval (Section 4.5). Nine participants rated the question count about right; P9 rated it 4 on the 1–5 scale, toward too many. Purchase-check means were 2.7 in Manual and 3.0 with PREFLOW over seven pairs (Table S20). The Agent midpoint mean hides variation: five chose 3, P10 chose 1 (far too few), and P9 chose 5 (far too many). In Manual, six chose 3, and P10 chose 1.

Reply length. Six rated replies about the right length, two somewhat too short (P2, P9), and two somewhat too long (P4, P8). Preferences for detail therefore varied; a single midpoint average would conceal them.

Control. Mean perceived control was slightly lower with PREFLOW (5.6 versus 5.9). Over seven paired respondents, clarity about the product, options, and price before buying was higher (6.7 versus 6.3). These measures cover different aspects of the experience. P1 wanted to shop independently with help available on demand; P9 preferred Manual for familiar products and the assistant for exploring unfamiliar ones. Together with the individual differences, this supports offering flexible assistance and direct browsing, rather than attributing the control difference to a single mechanism.

Product appearance. The seven paired respondents rated access to information about appearance higher with PREFLOW (5.7 versus 4.1; Table S20). P4 nevertheless discovered an unwanted style detail, cat ears on the selected coat, only at the end and requested full product titles and key appearance details. Their product-satisfaction rating was lower with PREFLOW (2 versus 6). Useful descriptions do not ensure that every decision-relevant detail is conveyed.

What participants valued. Positive feedback occurs in all ten participants' end-of-session answers (Table S21). P2 valued questions about features they had not considered. P3 said PREFLOW "really saves time, helps with choosing once the preferences are set, and helps a lot with comparing products" (translated from Arabic). P4–P8 described less listening or screen-reader navigation and easier filtering or budget changes. In the session recording, P9 valued filtering out irrelevant products, avoiding inspection of items that did not match their preferences. P10 described using the assistant as a good experience. These are reported experiences, not measured

time savings. Seven preferred PREFLOW and three had no overall preference; none preferred Manual overall.

S7 Beyond Shopping

In brief. The control flow and the six roles do not depend on shopping; the task schema, the evidence rules, and the checks before the final action do.

S7.1 What Changes for Another Task Family

Section 7 states that other long-horizon tasks need their own task schema, evidence rules, and checks for the final action, while the control flow and the roles stay the same. Table S22 makes this concrete for the traveler of Section 1, who rebooks a canceled flight while driving to the airport. The mapping is a design sketch; it is neither implemented nor evaluated.

S7.2 Summary

Across the three settings, the same pattern recurs. Where PREFLOW gains, it gains because the explicit memory keeps what the user wants and what the pages show outside the language model's context window: preferences survive long trajectories and a weaker LLM (Sections S4.2 and S4.3), questions target properties that the observed products actually have (Section S4.5), and the report can say what is in doubt before a costly action (Section S5). Where PREFLOW falls short—beliefs that misread a preference, matching rules designed for one shop, a study of ten people—the limitation is explicit and points to the next step.

References

[1] Amazon. 2026. Amazon's Holiday Kids Gift Book is back with 600+ toys, games, and gifts for every age. https://www.aboutamazon.com/news/retail/amazon-2026-holiday-kids-gift-book.

[2] Amazon. 2026. How to use Alexa for Shopping to compare products, check price history, auto-buy items at target prices, and more. https://www.aboutamazon.com/news/retail/how-to-use-amazon-shopping-ai-assistant.

[3] Amazon. 2026. Introducing Amazon's About You. https://www.aboutamazon.com/news/retail/amazon-about-you-personalization-preferences.

[4] Andrei Broder. 2002. A taxonomy of web search. *ACM Sigir forum* 36, 2 (2002).

[5] John Brooke. 1996. SUS: A 'Quick and Dirty' Usability Scale. In *Usability Evaluation in Industry*. doi:10.1201/9781498710411-35

[6] Cristiano Castelfranchi and Rino Falcone. 1998. Towards a Theory of Delegation for Agent-Based Systems. *Robotics Auton. Syst.* 24 (1998), 141–157. doi:10.1016/S0921-8890(98)00028-1

[7] Sanxing Chen, Sam Wiseman, and Bhuwan Dhingra. 2024. ChatShop: Interactive Information Seeking with Language Agents. arXiv:2404.09911 doi:10.48550/ARXIV.2404.09911

[8] Xiang Deng, Yu Gu, Boyuan Zheng, et al. 2023. Mind2Web: Towards a Generalist Agent for the Web. In *Proc. of NeurIPS '23.* doi:10.52202/075280-1220

[9] Zeyao Du, Tong Li, and Haibo Zhang. 2026. EComAgentBench: Benchmarking Shopping Agents on Long-Horizon Tasks with Distributed Hidden Intent. arXiv preprint arXiv:2606.17698. arXiv:2606.17698 doi:10.48550/arXiv.2606.17698

[10] Shicheng Fan, Mingdai Yang, Duohao Wang, et al. 2026. Agentic Commerce World: An Auditable and Verifiable Environment for Vibe Commerce. arXiv:2608.02441 [cs.AI] https://arxiv.org/abs/2608.02441

[11] Div Garg, Diego Caples, Andis Draguns, et al. 2025. REAL: Benchmarking Autonomous Agents on Deterministic Simulations of Real Websites. In *Proc. of NeurIPS '25.* doi:10.52202/085713-4536

[12] Google. 2025. AI Mode can now help you search and explore visually. https://blog.google/products-and-platforms/products/search/search-ai-updates-september-2025/.

[13] Google. 2025. Let AI do the hard parts of your holiday shopping. https://blog.google/products-and-platforms/products/shopping/agentic-checkout-holiday-ai-shopping/.

[14] Google. 2026. Get help with your shopping in Gemini Apps. Google Gemini Apps Help, https://support.google.com/gemini/answer/16730149.

[15] Boyu Gou, Zanming Huang, Yuting Ning, et al. 2025. Mind2Web 2: Evaluating Agentic Search with Agent-as-a-Judge. In *Proc. of NeurIPS '25.* doi:10.52202/085713-5778

[16] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Human Mental Workload*. Vol. 52. doi:10.1016/S0166-4115(08)62386-9

[17] InnoSearch AI. 2024. InnoSearch User Guide: Your Accessible Shopping Companion. https://blog.innosearch.ai/innosearch-user-guide-your-accessible-shopping-companion.

[18] InnoSearch AI. 2025. Introducing 1-855-SHOP-GPT (1-855-746-7478) 'Shop by Phone' Experience! https://blog.innosearch.ai/introducing-1-855-shop-gpt-1-855-746-7478-shop-by-phone-experience.

[19] Bernard J. Jansen, Danielle L. Booth, and Amanda Spink. 2008. Determining the informational, navigational, and transactional intent of Web queries. *Inf. Process. Manag.* 44, 3 (2008). doi:10.1016/J.IPM.2007.07.015

[20] Sunghwan Kim, Ryang Heo, Yongsik Seo, et al. 2026. AgenticShop: Benchmarking Agentic Product Curation for Personalized Web Shopping. In *Proc. of WWW '26.* doi:10.1145/3774904.3792724

[21] Serin Kim, Sangam Lee, and Dongha Lee. 2026. Persona2Web: Benchmarking Personalized Web Agents for Contextual Reasoning with User History. arXiv preprint. arXiv:2602.17003 https://arxiv.org/abs/2602.17003v1

[22] Xing Han Lù, Zdenek Kasner, and Siva Reddy. 2024. WebLINX: Real-World Website Navigation with Multi-Turn Dialogue. In *Proc. of ICML '24.* https://proceedings.mlr.press/v235/lu24e.html

[23] Kaixin Ma, Hongming Zhang, Hongwei Wang, et al. 2024. LASER: LLM Agent with State-Space Exploration for Web Navigation. arXiv preprint. Version 2, 21 February 2024. arXiv:2309.08172 doi:10.48550/arXiv.2309.08172

[24] OpenAI. 2025. Buy It in ChatGPT: Instant Checkout and the Agentic Commerce Protocol. https://openai.com/index/buy-it-in-chatgpt/.

[25] OpenAI. 2025. Introducing shopping research in ChatGPT. https://openai.com/index/chatgpt-shopping-research/.

[26] Ralph Peeters, Aaron Steiner, Luca Schwarz, et al. 2026. WebMall – A Multi-Shop Benchmark for Evaluating Web Agents. In *Proc. of SIGIR '26.* doi:10.1145/3805712.3808592

[27] Perplexity. 2025. Pro Perks, Academic homepage, and revamped Finance homepage. Perplexity Changelog, https://www.perplexity.ai/changelog/what-we-shipped-may-23rd. Includes the Shopping Follow-up Questions feature.

[28] Perplexity. 2026. What is Instant Buy? Perplexity Help Center, https://www.perplexity.ai/help-center/en/articles/10352906-what-is-instant-buy.

[29] Xavier Puig, Kevin Ra, Marko Boben, et al. 2018. VirtualHome: Simulating Household Activities via Programs. In *Proc. of CVPR '18.* doi:10.1109/CVPR.2018.00886

[30] Cheng Qian, Zuxin Liu, Akshara Prabhakar, et al. 2025. UserBench: An Interactive Gym Environment for User-Centric Agents. arXiv preprint arXiv:2507.22034. arXiv:2507.22034 [cs.AI] doi:10.48550/arXiv.2507.22034

[31] Jeff Sauro and Joseph S. Dumas. 2009. Comparison of Three One-Question, Post-Task Usability Questionnaires. In *Proc. of CHI '09.*

[32] Tianlin Shi, Andrej Karpathy, Linxi Fan, et al. 2017. World of Bits: An Open-Domain Platform for Web-Based Agents. In *Proc. of ICML '17.* http://proceedings.mlr.press/v70/shi17a.html

[33] Daniel Toyama, Philippe Hamel, Anita Gergely, et al. 2021. AndroidEnv: A Reinforcement Learning Platform for Android. arXiv preprint arXiv:2105.13231. arXiv:2105.13231 doi:10.48550/arXiv.2105.13231

[34] Shunyu Yao, Howard Chen, John Yang, et al. 2022. Webshop: Towards scalable real-world web interaction with grounded language agents. In *Proc. of NeurIPS '22.*

[35] Jingheng Ye, Yong Jiang, Xiaobin Wang, et al. 2025. ProductAgent: Benchmarking Conversational Product Search Agent with Asking Clarification Questions. In *Proc. of EMNLP '25 (Industry Track).* doi:10.18653/v1/2025.emnlp-industry.25

[36] Suyu Ye, Haojun Shi, Darren Shih, et al. 2026. RealWebAssist: A Benchmark for Long-Horizon Web Assistance with Real-World Users. In *Proc. of AAAI '26.* doi:10.1609/AAAI.V40I40.40742

[37] Shuyan Zhou, Frank F. Xu, Hao Zhu, et al. 2024. WebArena: A Realistic Web Environment for Building Autonomous Agents. In *Proc. of ICLR '24.* https://openreview.net/forum?id=oKn9c6ytLx

Table S21: End-of-session feedback and overall condition preference. Direct participant quotations are italicized; improvement requests are paraphrased. P3's quotation was translated from Arabic. Favorable experiences coexist with requests for improvement.

Participant	Preferred	Experience or comment
P1	No preference	*"Product descriptions were very helpful"* Wanted to browse independently with assistance available when needed.
P2	PREFLOW	*"I liked the questions about more features that I may not have thought about including in my search."*
P3	PREFLOW	*"Indeed, the assistant saves time, helps with choosing when preferences are specified, and helps a lot with comparing products."*
P4	PREFLOW	*"The assistant made finding items and adjusting to the new budget super fast and cut down a lot of NVDA navigation effort."*
P5	PREFLOW	*"The shopping assistant worked very well for summarizing relevant options and quickly adjusting the search when my budget changed."*
P6	PREFLOW	*"It made switching budgets and choosing options much less tiring than browsing on my own."*
P7	PREFLOW	*"It gathered the details about size, price, and features directly, which made the whole process much simpler."*
P8	PREFLOW	*"Having the assistant check colors and sizes directly saved me a lot of voiceover navigation."* Wanted shorter replies and fewer initial questions.
P9	No preference	*"The assistant describes the products very well, helped me a lot to discover unfamiliar products. The assistant was also good at filtering out the noise; I didn't have to look into products that don't match what I want"* Preferred Manual for familiar purchases and the assistant for discovery, with richer information during search.
P10	No preference	*"Good experience using it."*

Table S22: What stays and what changes when PREFLOW moves from shopping to flight rebooking. The rebooking column is a design sketch, not an implemented or evaluated system.

Element	Shopping (evaluated)	Flight rebooking (sketch)
Task schema σ	transactional.obtain: product; attributes; options; other preferences	transactional.obtain: flight; attributes such as arrival time, stops, airline; options such as seat and fare class; other preferences such as price and refundability
Preference sources	Instruction, profile, replies	Instruction ("get me home tonight"), profile (seat preference, loyalty program), replies
Evidence for the Evidence Ledger	Results page, product page, option controls, reviews	Flight results, fare rules, seat map
A typical unresolved preference	How hard a desk is to assemble	Whether checked baggage is transferred
Check before the final action	Product type, price, attributes, options	Arrival time, fare difference, change fees, connection times
Dialogue	Screen-reader announcements; purchase summary with product, options, price, and unmet preferences	Short spoken messages; questions timed to the driver's attention; purchase summary with flight, fare, and unmet preferences
Unchanged	Algorithm 1; the six roles; the statuses supported, contradicted, and unresolved; approval before an action that is costly to reverse; the report of unmet preferences.	